

By TOUCH DEN · November 12, 2026

$ Why Security People Use Linux

Video 1 · What Linux Is, and Where It Runs

Linux for Security · Section 1

In this video, you will learn:

- What Linux is
- Why open source matters for security
- Where Linux runs, and why that matters

What Linux is

Linux is an operating system, like Windows or macOS (course 02): it manages the hardware and runs programs.

- The name refers to the **kernel** (course 02's privileged core), written by Linus Torvalds in 1991, built on by thousands since.
- Two differences that matter for security: it is **free and open source**, and it is built around the **command line**.

Rare on home laptops; **everywhere behind the scenes** of the internet.

Open source, and why it matters

Open source = the code is public; anyone can read, check, and change it. (The opposite of Windows' secret code.)

- **Anyone can inspect it:** flaws and backdoors are harder to hide; you can **verify**, not just trust.
- **You can see exactly what it does**, priceless for a security person.
- **Free and modifiable:** the whole security-tool ecosystem grew from it.

A field built on **understanding systems** prefers a system you're allowed to understand.

Where Linux runs

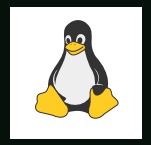
Where	How much Linux
Web servers	Most of the world's servers
The cloud	Most cloud computers
Phones	Android (billions) is built on Linux
Devices (IoT)	Routers, TVs, cameras, cars
Supercomputers	Essentially all

The systems attackers target and defenders protect are mostly Linux. Can't work on Linux → locked out of real security work.

Recap: what Linux is

- Linux is a **free, open-source OS**; its kernel runs the systems that matter.
- **Open source** lets you verify, not just trust, it fits security's spirit.
- Most servers, the cloud, Android, and devices are Linux, the real targets and defenses.

Next video: security tools, distributions, and the command-line mindset.



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$ Why Security People Use Linux

Video 2 · Tools, Distros, and the Command Line

Linux for Security · Section 1

In this video, you will learn:

- Why security tools live on Linux
- Distributions, and which to use
- The command-line mindset

Why security tools live on Linux

Most security software is built for Linux, and much runs **only** on Linux.

- **Deep control:** low-level access to packets, processes, files.
- **Scriptable:** combine tools and automate (the heart of efficient work).
- **Free:** run dozens of lab machines, no licensing.
- **The community builds there**, new tools appear on Linux first.

There's even **Kali Linux**: a distro pre-loaded with offensive tools (your future "attacker machine").

Distributions

No single "Linux" to download. Groups package the kernel + programs into a **distribution** (distro). Same kernel → skills transfer.

Distro	Known for
Ubuntu	Friendly, popular, well-documented, **this course**
Debian	Stable, minimal (Ubuntu is built on it)
Kali	Offensive security tools (the attacker's distro)
Red Hat / Rocky	Enterprise servers

Learn on **Ubuntu**; skills carry to all of them.

The command-line mindset

On Linux, security work is done by **typing commands**, not clicking. Hard at first, then far faster and more powerful.

- **Precise:** does exactly what you say.
- **Powerful:** pipelines do more than clicks (lesson 8).
- **Repeatable/automatable:** save in a script, run anywhere.
- **Works everywhere:** servers have **no mouse**, you reach them by command (SSH).

Expect it to feel slow, then click, then clicking a mouse feels clumsy.

Recap: tools, distros, command line

- Almost all **security tools** live on Linux (including Kali).
- A **distribution** packages Linux; this course uses **Ubuntu**; skills transfer.
- The **command line** is precise, powerful, automatable, and the only option on servers, embrace it.

Next lesson: install Linux in a virtual machine, your safe practice system.